MID SEMESTER EXAMINATION - 1, THEORY EVEN SEM - 2025-26

CA105B - ADVANCED JAVA PROGRAMMING

Programme Name: MCA **Semester: 2**

Max. Marks: 30 **Max. Time: 2 hours** **Question Paper ID: 001304**

Instructions:
All questions are compulsory.
Assume suitable data wherever necessary and mention it clearly.

Section-1 (5 × 6 = 30 Marks)

		Marks	CO	BL

Q1. a)	**Predict the output and justify your answer.**
	class A {
	void show() { System.out.println("A"); }
	}
	
	class B extends A {
	void show() { System.out.println("B"); }
	}
	
	public class Test {
	public static void main(String[] args) {
	A obj = new B();
	obj.show();
	}
	}
		2	1	2

b)	**Predict the output and justify your answer.**
	class A {
	static int x = 100;
	int y = 200;
	}
	
	public class Demo {
	public static void main(String[] args) {
	A a1 = new A();
	A a2 = new A();
	a1.x = 500;
	a2.y = 700;
	System.out.println(a2.x + " " + a1.y);
	}
	}
		2	1	2

c)	**Which of the following supports runtime polymorphism?** a) Method overloading b) Method overriding c) Constructor overloading d) Static methods **Choose the correct option and define that concept.**	2	1	1

Q2. **Answer any 2 Question(s)**

a)	Describe the use of super() method with the help of an example.	3	1	2
b)	Explain the role of constructors in Java. Also explain default and parameterized constructors with examples.	3	1	2
c)	Illustrate the concept of inheritance and its types with the help of diagrams.	3	1	3
Q3. a)	Demonstrate the use of a final class in Java by writing a suitable example.	3	1	3

	b)	Compare abstract classes and interfaces with example.	3	1	2
Q4.	**a)**	Can a functional interface have static methods? Explain with help of an example.	3	2	2
	b)	**Syntax of lambda expression is:** a) (parameters) → expression b) (parameters) ; expression c) parameters = λ expression **Choose the correct option. Also write an example with lambda expression.**	3	2	1

Q5. **Answer any 2 Question(s)**

	a)	Explain the primary purpose of the Java Module System (JPMS) ?	3	2	2
	b)	Explain the different types of method references with syntax.	3	2	2

	c)	**Predict the output and justify your answer.**
		int num = 4;
		String ans = switch(num) {
		case 1 -> "One";
		case 2 -> "Two";
		default -> {
		yield "Other Number";
		}
		};
		System.out.println(ans);
			3	2	2

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